



KUBERNETES ROADMAP

Step-by-Step Guide to Become a Kubernetes Expert



LEARN → PRACTICE → BUILD → DEPLOY → MASTER

1		1. Prerequisites - Linux Fundamentals - Networking Basics - Basic understanding of Docker - CLI & Command Line Skills	1–2 Weeks Resources Linux Journey, Docker Docs
2		2. Containerization with Docker - Images, Containers, Volumes - Dockerfile, Build, Run, Push - Docker Compose	1–2 Weeks Resources Docker Docs, Play with Docker
3		3. Kubernetes Fundamentals - What is Kubernetes? - Architecture & Components - Kubernetes Cluster Setup (kubeadm) - kubectl Basics	2–3 Weeks Resources Kubernetes Docs, Kubernetes the Hard Way
4		4. Core Kubernetes Concepts - Pods, ReplicaSets, Deployments - Services & Networking - ConfigMaps, Secrets - Namespaces, Labels & Annotations	2–3 Weeks Resources Kubernetes Docs, KodeKloud, Playgrounds
5		5. Storage in Kubernetes - Volumes & VolumeMounts - Persistent Volumes (PV & PVC) - Storage Classes	1–2 Weeks Resources Kubernetes Docs
6		6. Networking in Kubernetes - Cluster Networking Basics - Services (ClusterIP, NodePort, LoadBalancer) - Ingress & Ingress Controller - Network Policies	1–2 Weeks Resources Kubernetes Docs, Cilium/Calico Docs
7		7. Security in Kubernetes - RBAC (Roles, RoleBinding, ServiceAccounts) - Pod Security Standards - Secrets Management - Network Policies	1–2 Weeks Resources Kubernetes Docs
8		8. Observability & Troubleshooting - Liveness, Readiness & Startup Probes - Logging (ELK / EFK Stack) - Monitoring (Prometheus & Grafana) - kubectl Debugging, Events	1–2 Weeks Resources Prometheus Docs, Grafana Labs
9		9. Helm Charts - What is Helm? - Helm Install, Upgrade, Rollback - Create Your Own Helm Chart	1 Week Resources Helm Docs
10		10. CI/CD with Kubernetes - GitOps Concepts - CI/CD Tools: Jenkins / GitHub Actions / GitLab CI - Argo CD / Flux	2 Weeks Resources ArgoCD Docs, Flux Docs
11		11. Advanced Topics - StatefulSets, DaemonSets, Jobs, CronJobs - Horizontal Pod Autoscaler (HPA) - Resource Limits & Quotas - Operators (Basics)	2 Weeks Resources Kubernetes Docs
12		12. Projects & Real World Practice - Deploy a Multi-tier Application - Setup Monitoring & Logging - Implement CI/CD Pipeline - Use Helm for Deployment	Ongoing Resources Hands-on Projects, Your Cloud / Lab

ESSENTIAL SKILLS

- Container Orchestration
- Linux Administration
- Networking
- Automation & Scripting
- Problem Solving
- DevOps Mindset

IMPORTANT KUBERNETES COMPONENTS

- | | |
|--------------------|-------------------|
| API Server | kubelet |
| etcd | kube-proxy |
| Scheduler | CoreDNS |
| Controller Manager | Container Runtime |

TOOLS TO KNOW

- | | | |
|---------|------------|---------|
| kubectl | Helm | K9s |
| Lens | Prometheus | Grafana |
| Git | Docker | Harbor |

LEARNING PATH FLOW



BEST PRACTICES

- ✓ Start Small and Keep Practicing
- ✓ Automate Everything
- ✓ Use Infrastructure as Code
- ✓ Monitor and Observe Your Cluster
- ✓ Keep Learning & Stay Updated
- ✓ Join Kubernetes Community

CAREER OPPORTUNITIES

- DevOps Engineer
- Kubernetes Administrator
- Cloud Engineer
- Site Reliability Engineer (SRE)
- Platform Engineer



Consistency + Practice + Real Projects = Kubernetes Mastery

